Electric fields

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use field arrows to predict the force on a test charge.

SUCCESS CRITERIA - I can...

1. Draw electric field arrows for isolated positive and negative charges.
2. Use the positive-test-charge convention.
3. Distinguish the field from a particular test charge's force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "electric field" and "charge" to explain your model or result. Sentence starters are optional.

electric field: force per positive test charge at a location

charge: a property responsible for electric interactions

vector: a quantity with size and direction

01 Bring back an earlier idea

A ray enters a medium with speed 1/3 of its original speed at an angle.
Sketch the bend and state whether frequency changes.

Word help: frequency = number of cycles each second

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: E = F/q; F = qE.
Use consistent signed components. The field points along the force on a positive test charge; negative charge reverses force direction.

Worked reasoning: At a point where E=4 N/C right, a+2 C test charge feels 8 N right. A-2 C charge feels 8 N left. Field direction is defined using positive test charge.

Your turn: Draw the field arrow once and the two alternative force arrows.

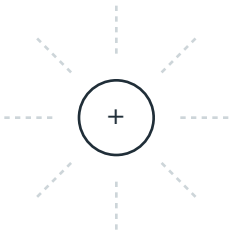
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Look and explain

Sketch field arrows around a positive source charge. Place a negative test charge to its right and draw that charge's force.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Add arrowheads to show field direction.

The center marks the source; add arrowheads on the surrounding lines to show field direction.

Marzano check - circle: ① ② ③

04 Build the solution

A positive 2 C test charge feels 8 N right.
Find E=F÷q and its direction.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

If you remove the test charge, does the source charge's field disappear?
Explain the role of a test charge.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

06 Choose a method

An echo travels 120 m on its full out-and-back path.
Find the wall's distance.

Word help: echo = a sound received after reflection

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

07 Exit check

A field points down. Which way is the force on a positive charge? On a negative charge?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

08 STRETCH: Apply it another way

Return to the worked example in problem 2.
Choose two different charge values giving forces with equal magnitudes and opposite directions in the same field.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③